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Table 13.6: Difference between photosynthesis in plants and photosynthesis in bacteria	
Photosynthesis in Plants	**Photosynthesis in Bacteria**
1. Cyclic and non-cyclic phosphorylation takes place	1. Only cyclic phosphorylation takes place
2. Photosystem I and II involved	2. Photosystem I only involved
3. Electron donor is water	3. Electron donor is H_2S
4. Oxygen is evolved	4. Oxygen is not evolved
5. Reaction centres are P700 and P680	5. Reaction centre is P_{870}
6. Reducing agent is NADPH + H^+	6. Reducing agent is NADH + H^+
7. PAR is 400 to 700 nm	7. PAR is above 700 nm
8. Chlorophyll, carotenoid and xanthophyll	8. Bacterio chlorophyll and bacterio viridin
9. Photosynthetic apparatus – chloroplast	9. It is chlorosomes and chromatophores

13.18 Photosynthesis in bacteria

Though we study about bacterial photosynthesis as the last part, bacterial photosynthesis formed first and foremost in evolution. Bacteria does not have specialized structures like chloroplast. It has a simple type of photosynthetic apparatus called chlorosomes and chromatophores (Table 13.6). **Van Neil** (1930) discovered a bacterium that releases sulphur instead of oxygen during photosynthesis. Here, electron donor is hydrogen sulphide (H_2S) and only one photosystem is involved (PS I) and the reaction centre is P_{870}. Pigments present in bacteria are bacteriochlorophyll a, b, c, d, e and g and carotenoids. Photosynthetic bacteria are classified into three groups:

1. Green sulphur bacteria. Example: *Chlorobacterium* and *Chlorobium*.
2. Purple sulphur bacteria. Example: *Thiospirillum* and *Chromatium*.
3. Purple non-sulphur bacteria. Example: *Rhodopseudomonas* and *Rhodospirillum*.

Summary

Photosynthesis is an oxidation and reduction process. It has two phases: the light reaction and dark reaction. During light reaction water is oxidised to release O_2 and during dark reaction $CO_2$is reduced to form sugars. Solar energy is trapped by pigment system I and pigment system II. P700 and P680 act as reaction centres for PS I and PS II respectively. Splitting of water molecule (Photolysis) produces electrons, protons and oxygen. Photophosphorylation takes place through cyclic and non-cyclic mechanisms and generates energy and reducing power. Dark reaction or biosynthetic phase of photosynthesis use the products of light energy (ATP and NADPH + H^+) and carbon dioxide is reduced to Carbohydrates. Carbon pathway in C_3 cycle has RUBP as the acceptor molecule and the first product is PGA (3C). Carbon pathway in C_4 plants involves mesophyll and bundle sheath cells, Kranz anatomy. Dimorphic chloroplast, no photorespiration, acceptor molecule as PEP and first product as OAA (4C) are some of the unique characters of C_4 cycle. C_2 Cycle or photorespiration is operated when less amount of CO_2 is used for reduction and O_2 increases. Rubisco starts to